Feedback Control Design Process with Performance Specifications

(性能指定的反馈控制设计过程)

Feedback Control Design Process

(1) Model the physical system (plant) that we want to control and obtain its I/O transfer function $P(s)$ **(对被控物理系统建模并获得输入输出传递函数).**

(2) Determine sensor dynamics and actuator dynamics (if necessary) **(确定传感器动力学和执行器动力学).**

(3) Draw the closed-loop block diagram, which includes the plant, sensor, actuator and controller transfer functions $C(s)$ and $F(s)$ **(画出闭环系统方框图).**

(4) Obtain the closed-loop transfer functions **(获得闭环传递函数).**

(5) Based on the performance specifications, find the conditions that the CLTFs have to satisfy **(基于指定的性能，找出闭环传递函数满足的约束条件).**

(6) Choose controller structures for $C(s)$ and $F(s)$ and substitute them into the CLTFs **(选择控制器结构并将其代入闭环传递函数).**

(7) Select the controller parameters so that the design constraints are satisfied **(选择控制器参数以使约束条件满足).**

(8) Verify your design via computer simulation and actual implementation **(通过计算机仿真和实际实施来验证设计).**

In Class Exercise

You are the young engineer that is in charge of designing the control system for the next generation inkjet printer (refer the example discussed in previous lecture notes). During the latest design review, the following plant parameters are obtained **(喷墨打印机的控制系统参数如下):**

L_A = 10 mH
R_A = 10 Ω
K_T = 0.06 Nm/A
$J_E = 6.5 \times 10^{-6}$ Kg m^2
$B_E = 1.4 \times 10^{-5}$ Nm/(rad/sec)

The drive roller angular position is sensed by a rotational potentiometer with a static sensitivity of K_S = 0.03 V/deg **(主动辊转角位置由静态灵敏度系数为K_S = 0.03 V/deg的旋转电位器来测量).**

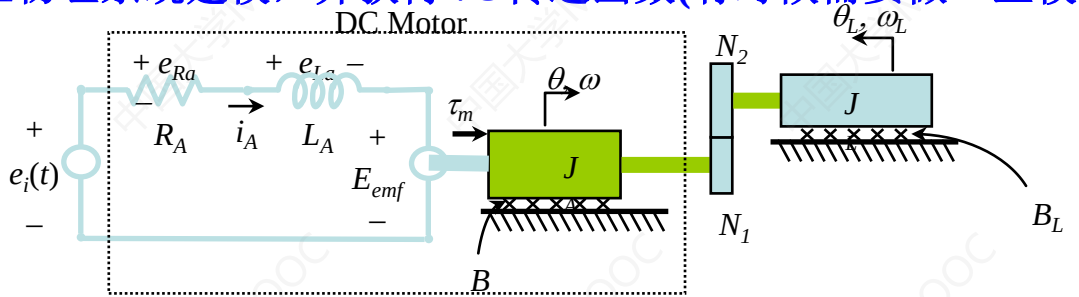
The design specifications for the paper positioning system are **(纸张定位系统的设计要求指定为):**

- The steady state position for a step input should be within 5% of the desired position **(阶跃输入的稳态位置在期望位置的5%范围内).**
- The 2% settling time should be less than 200 msec, and **(2%调整时间小于200毫秒)**
- the percent overshoot should be less than 5% **(超调量小于5%).**

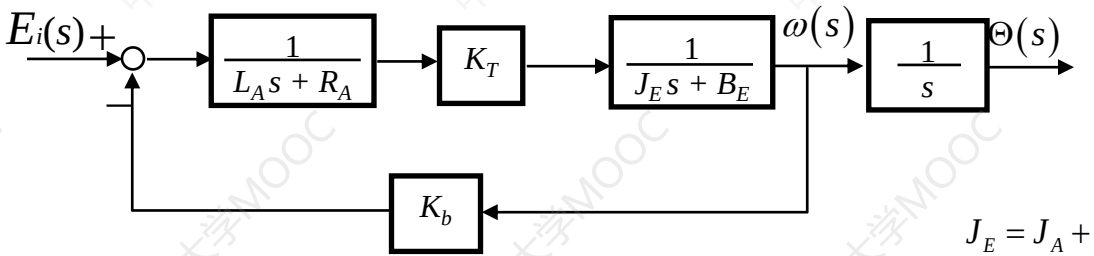
You are to design a controller that satisfies the above specifications **(设计控制器满足上述要求).**

In Class Exercise

(1) Model the physical system (plant) that we want to control and obtain its I/O transfer function $G(s)$. (Sometimes, certain model simplification should be performed.) **(对被控物理系统建模，并获得I/O传递函数(有时候需要做一些模型简化))**



From previous example, the DC motor driven paper positioning system can be modeled by **(直流电机驱动的打印纸定位系统可建模成)**



$$P(s) = \frac{\Theta(s)}{E_i(s)}$$

$$J_E = J_A + \left(\frac{1}{N}\right)^2 \cdot J_L, \quad B_E = B + \left(\frac{1}{N}\right)^2 \cdot B_L, \quad N = \frac{N_2}{N_1}$$

In Class Exercise

The plant transfer function P(s) can be derived to be **(推导系统传递函数)**:

$$P(s)=\frac{\theta(s)}{E_i(s)}=\frac{K_T}{s(L_AJ_Es^2+(B_EL_A+R_AJ_E)s+(R_AB_E+K_bK_T))}$$

Simplify the plant model by neglecting the electrical subsystem dynamics (i.e., by letting $L_A = 0$) **(通过忽略电路子系统的动力学来简化系统模型)**:

$$P(s)=\frac{\theta(s)}{E_i(s)}=\frac{K_T}{s(R_AJ_Es+(R_AB_E+K_bK_T))}=\frac{\underbrace{\frac{K_T}{R_AB_E+K_bK_T}}_{K_M}}{s(\underbrace{\frac{R_AJ_E}{R_AB_E+K_bK_T}}_{\tau_M}s+1)}=\frac{K_M}{s(\tau_Ms+1)}$$

Substituting in the numerical values, we have our plant transfer function **(代入数值，得到系统的传递函数):**

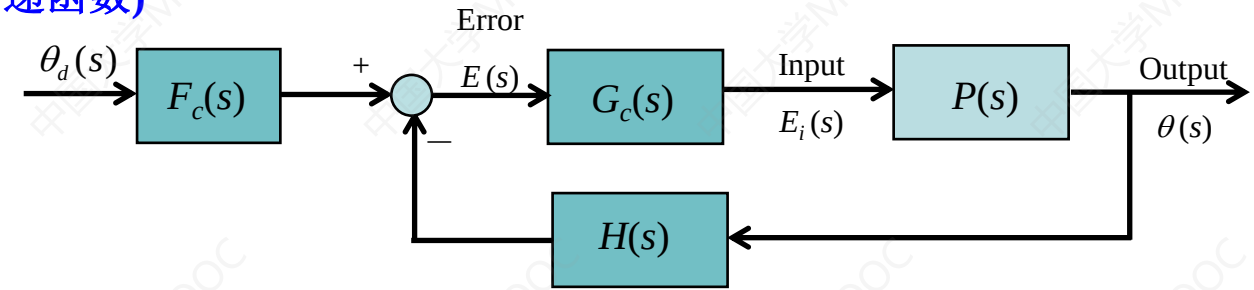
$$P(s)=\frac{K_M}{s(\tau_Ms+1)}=\frac{16}{s(0.017s+1)}$$

(2) Determine sensor dynamics (transfer function of the measurement system) H(s) and actuator dynamics (if necessary) **(确定传感器动力学(测量系统的传递函数)和执行器动力学(如果必要))**.

$$H(s)=0.03\text{V/deg}=1.719\text{V/rad}$$

In Class Exercise

(3)Draw the closed-loop block diagram, which includes the plant, sensor, actuator and controller $G_C(s)$ transfer functions **(画出闭环方框图，包括系统，传感器，执行器和控制器的传递函数)**



(4) Obtain the closed-loop transfer functions **(获得闭环传递函数)**

$$G_{YR}(s)=\frac{\theta(s)}{\theta_d(s)}=\frac{G_c(s)P(s)F(s)}{1+G_c(s)P(s)H(s)}=\frac{1}{H(s)}\frac{G_c(s)P(s)H(s)F(s)}{1+G_c(s)P(s)H(s)}$$

$$F(s)=H(s)=1.719$$

In Class Exercise

(5) Based on the performance specifications, find the conditions that $T_r(s)$ has to satisfy
(基于指定的性能，找出$T_r(s)$应该满足的条件)

Steady State specification **(稳态性能)**:

System type 1 $\quad 0.95 \le G_{YR}(0) = 1 \le 1.05$

Transient Specifications **(瞬态性能)**

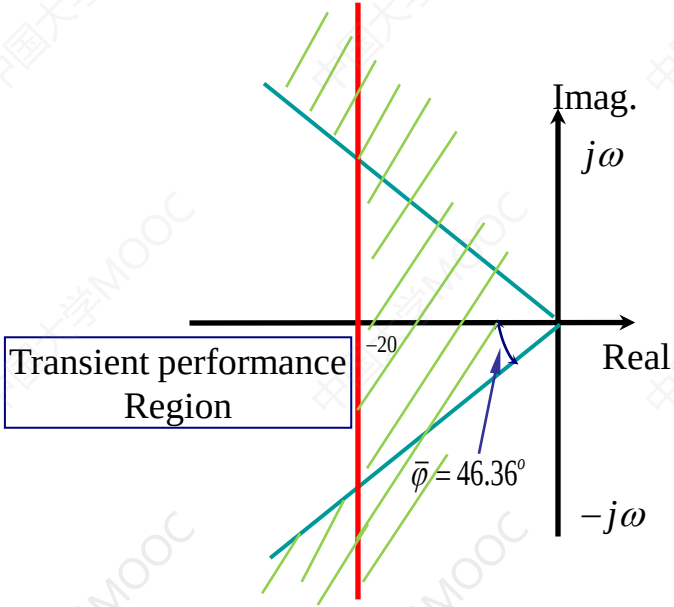
Settling Time Constraint:

$$t_s \le 0.2\,\text{sec} \Leftrightarrow \sigma \ge \frac{4}{t_s} = 20$$

Overshoot Constraint:

$$OS\% \le 5\% \quad \Rightarrow \quad \varphi \le \bar{\varphi} = \arctan\left(-\frac{\pi}{\ln(X\%)}\right)$$

$$= \tan^{-1}\left(-\frac{\pi}{\ln(0.05)}\right) = 46.36^{\circ}$$



In Class Exercise

(6) Choose controller structure $C(s)$ and $F(s)$, and obtain the CLTFs **(选择控制器结构$C(s)$和$F(s)$,并获得闭环传递函数)**.

$$P(s)=\frac{K_M}{s(\tau_M s+1)}=\frac{16}{s(0.017s+1)}$$

$$H(s)=1.719\text{V/rad}$$

Let's try a simple proportional control **(选择简单比例控制)**:

$$e_i(t)=K_P e_{\theta V}(t)=K_P(\theta_{dV}(t)-\theta_V(t))$$

In s-domain (Laplace domain), this control law can be written as **(s域内的控制率可写成):**

$$G_c(s)=K_p \qquad E_i(s)=K_P E(s)$$

Substitute the controller transfer function into $T_r(s)$ **(将控制器传递函数代入$T_r(s)$)**:

$$T_r(s)=G_{YR}(s)=\frac{\theta(s)}{\theta_d(s)}=\frac{G_c(s)P(s)F(s)}{1+G_c(s)P(s)H(s)}=\frac{1}{H(s)}\frac{G_c(s)P(s)F(s)H(s)}{1+G_c(s)P(s)H(s)}$$

$$=\frac{F(s)}{H(s)}\frac{K_p\dfrac{16}{s(0.017s+1)}1.719}{1+K_p\dfrac{16}{s(0.017s+1)}1.719}=\frac{F(s)}{H(s)}\frac{27.504K_p}{0.017s^2+s+27.504K_p}$$

In Class Exercise

(7) Select the controller parameters to satisfy the design constraints established in (5) **(选择控制器参数以满足式(5)所建立的约束条件)**

Steady State Constraint **(稳态约束)**:

Want: $0.95 \le T_r(0) \le 1.05$ *Actual:* $0.95 \le T_r(0) = \left| \frac{F(s)}{H(s)} \frac{27.504K_p}{27.504K_p} \right| = \frac{|F(s)|}{1.719} = 1 \le 1.05$

Transient Constraints **(瞬态约束)**:

Choose K_P such that the closed-loop poles are within the allowable region on the complex plane. Find an expression for the closed-loop poles **(选择K_p以使闭环极点在复平面允许的范围内，获得闭环极点的表达式)**:

$$\text{CLCE}: s^2 + 57.47s + 1580.5K_p = 0$$

$$\Rightarrow \text{Two CL poles} \quad p_{1,2} = -28.7 \pm \sqrt{826 - 1580.5K_p}$$

$$\Rightarrow \text{when } K_p > 0.5226,\ p_{1,2} = -28.7 \pm j\sqrt{1580.5K_p - 826}$$

$$\therefore \quad \frac{\sqrt{1580.5K_p - 826}}{28.7} = \tan(\varphi) < \tan(\bar{\varphi}) = \tan(46.36^\circ) \quad \Rightarrow \quad K_p < 1.0957$$

$$\text{when } K_p \le 0.5226,\ p_{1,2} = -28.7 \pm \sqrt{826 - 1580.5K_p}$$

$$\therefore \quad -28.7 + \sqrt{826 - 1580.5K_p} \le -20 \quad \Rightarrow \quad K_p \ge 0.4747$$

In Class Exercise

For every K_P , there will be two closed-loop poles. It is obvious that the two closed-loop poles change with the selection of different K_P **(对每个K_p有两个闭环极点,很明显两个闭环极点随着K_p的不同选择而变化).** For example:

$K_P = 0 \rightarrow p_{1,2} = 0, -57.47$

$K_P = 0.26 \rightarrow p_{1,2} = -8.4, -50$

$K_P = 0.475 \rightarrow p_{1,2} = -20.1, -37.4$

$K_P = 0.52 \rightarrow p_{1,2} = -28.7, -28.7$

$K_P = 0.7 \rightarrow p_{1,2} = -28.7 \pm 17j$

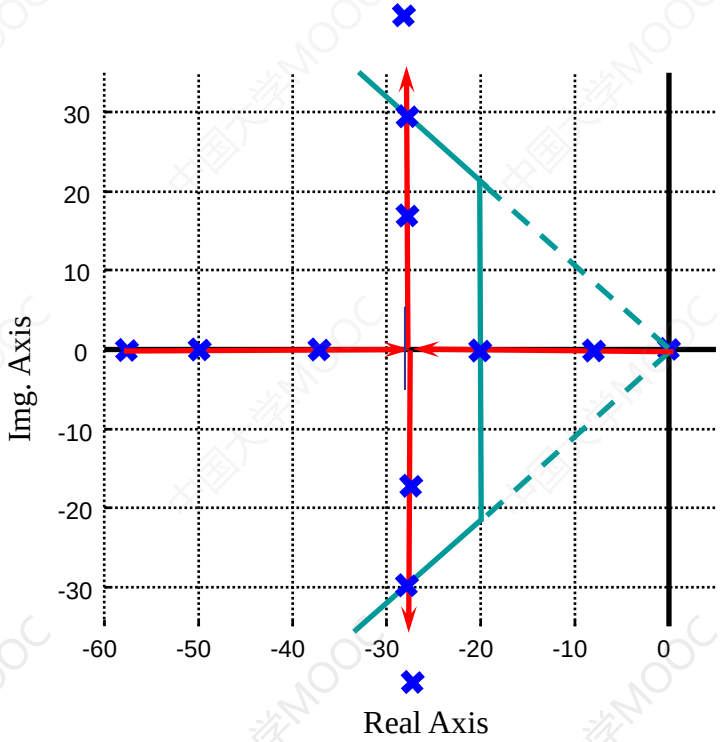
$K_P = 1.08 \rightarrow p_{1,2} = -28.7 \pm 29.7j$

$K_P = 1.75 \rightarrow p_{1,2} = -28.7 \pm 44j$

By inspecting the root-locus, we can find that if **(通过根轨迹检查，我们发现如果)**

$$1.08 \geq K_p \geq 0.475$$

then the closed-loop poles will be in the allowable region and the performance specifications will be satisfied **(闭环系统极点将在允许范围内，指定的性能得到满足)**.

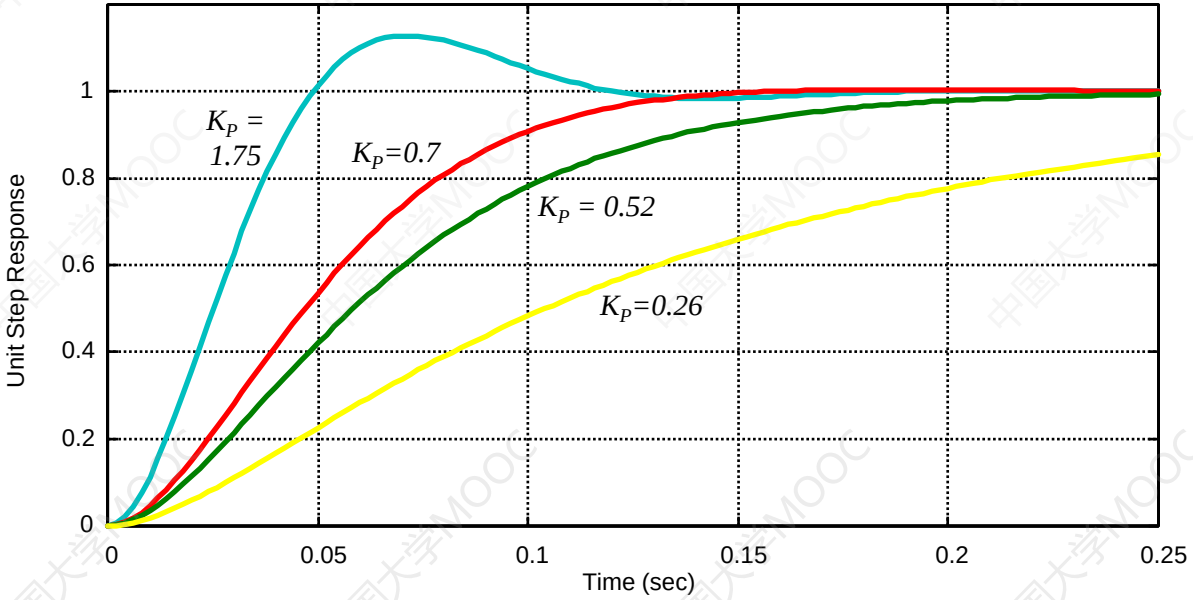


In Class Exercise

(8)Verify your design via computer simulation and actual implementation(通过仿真和实际实施来验证设计)

```
>> num = 16*Ks*Kp;
>> den = [tauM 1 16*Ks*Kp];
>> T = (0:0.0002:0.25)';
>> y = step(num,den,T);
>> plot(T,y);
```

$0.95<y_{ss}<1.05$
2% band: <0.2s
OS% : <5%



Summary

- **Steady State Performance Specifications (稳态性能指定)**
  - Steady-State Errors (稳态误差)
  - Following Polynomial Inputs and System Type (time-domain method) (跟踪多项式形式的输入和系统类型(时域方法))
  - Following Sinusoidal Inputs at Low Frequency (frequency domain method) (在低频段跟踪正弦形式输入(频域方法))

- **Transient Performance Specifications (瞬态性能指定)**
  - Unit Step Response (单位阶跃响应)
  - Second-order Equivalence Approximation (dominant poles concept) (二阶等效近似)